

21.1 Useful formulas

i. Given $A = \begin{bmatrix} a & b \\ b & c \end{bmatrix}$ and $\vec{x} = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$

$$\langle \vec{x}, A\vec{x} \rangle = \vec{x}^T A \vec{x} = ax_1^2 + 2bx_1x_2 + cx_2^2$$

ii. $\nabla \left(\frac{1}{2} \vec{x}^T A \vec{x} - \vec{x}^T \vec{b} \right) = A\vec{x} - \vec{b}$

Exercise

1) Find the minimum value of

$$\lambda^2 - 8\lambda + 11 = 0$$

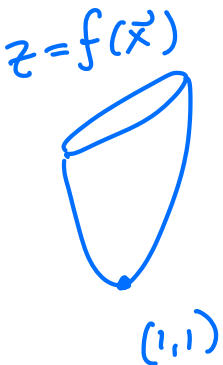
$$(\lambda - 4)^2 - 5 = 0$$

$$\lambda = 4 \pm \sqrt{5}$$

$$f = \frac{1}{2} \vec{x}^T A \vec{x} - \vec{x}^T \vec{b} + 4$$

where $A = \begin{bmatrix} 3 & -2 \\ -2 & 5 \end{bmatrix}$ and $\vec{b} = \begin{bmatrix} 1 \\ 3 \end{bmatrix}$

using linear algebra and justify why your value is minimum.



Since A has e-values of $\lambda = 4 \pm \sqrt{5}$ which are both positive, A is pos. def. and thus f has minimum at its vertex. Now setting $\nabla f = A\vec{x} - \vec{b}$

we have that

$$\begin{aligned}\vec{x} &= A^{-1} \vec{b} = \frac{1}{11} \begin{bmatrix} 5 & 2 \\ 2 & 3 \end{bmatrix} \begin{bmatrix} 1 \\ 3 \end{bmatrix} \\ &= \frac{1}{11} \begin{bmatrix} 11 \\ 11 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}\end{aligned}$$

and thus the minimum of f is

$$\begin{aligned}f(1,1) &= \frac{1}{2} \begin{bmatrix} 1 & 1 \end{bmatrix} \underbrace{\begin{bmatrix} 1 \\ 3 \end{bmatrix}}_{A \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 1 \\ 3 \end{bmatrix}} - \begin{bmatrix} 1 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 3 \end{bmatrix} + 4 \\ &= 2\end{aligned}$$

The equation $\vec{x}^T A \vec{x} = 1$ and Conic sections

Let $A \in \mathbb{R}^{2 \times 2}$ be pos. def. such that

$A = Q \Lambda Q^T$ where Λ is diagonal and Q is orthogonal. Then

$$\begin{aligned}\vec{x}^T A \vec{x} &= \vec{x}^T Q \Lambda Q^T \vec{x} \\ &= (Q^T \vec{x})^T \Lambda (Q^T \vec{x}) \\ &= \vec{y}^T \Lambda \vec{y}\end{aligned}$$

$$= \lambda_1 \gamma_1^2 + \lambda_2 \gamma_2^2$$

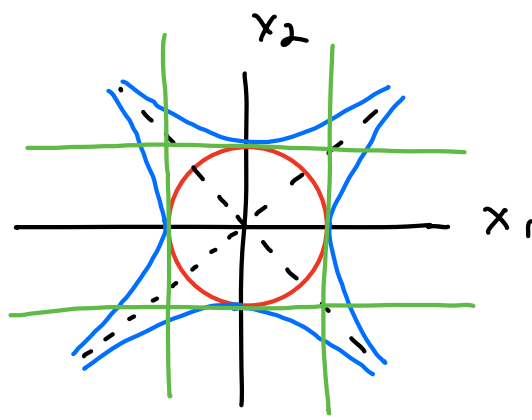
where $\begin{bmatrix} \gamma_1 \\ \gamma_2 \end{bmatrix} = Q^T \vec{x}$. Now we have that

Theorem 21.2 Given symmetric $A \in \mathbb{R}^{2 \times 2}$ and orthonormal $\vec{q}_1$ and $\vec{q}_2$ s.t.

$$A = \lambda_1 \vec{q}_1 \vec{q}_1^T + \lambda_2 \vec{q}_2 \vec{q}_2^T \quad (= Q \Lambda Q^T)$$

we have that if

- i. $\lambda_1 = \lambda_2 > 0$, then $\vec{x}^T A \vec{x} = 1$ is a circle (radius is $1/\sqrt{\lambda}$)
- ii. $\lambda_1 > \lambda_2 > 0$, then $\vec{x}^T A \vec{x} = 1$ is an ellipse (half lengths are $1/\sqrt{\lambda_i}$)
- iii. $\lambda_1 > 0 > \lambda_2$, then $\vec{x}^T A \vec{x} = 1$ is a hyperbola
- iv. $\lambda_1 > 0$ and $\lambda_2 = 0$, then $\vec{x}^T A \vec{x} = 1$ is two parallel lines
- v. $\lambda_1 \leq \lambda_2 \leq 0$, then $\vec{x}^T A \vec{x} = 1$ has no solutions (in $\mathbb{R}$)



$$x_1^2 + x_2^2 = 1$$

$$x_1^2 - x_2^2 = 1$$

$$x_1^2 = 1$$

$$x_2^2 = 1$$

Exercise:

2) Sketch the following conic section

$$a. \quad \overset{a}{11}x_1^2 - \overset{2b}{6}x_1x_2 + \overset{c}{3}x_2^2 = 1$$

Setting $A = \begin{bmatrix} 11 & -3 \\ -3 & 3 \end{bmatrix}$ so that the

given conic section is of the form

$\vec{x}^T A \vec{x} = 1$ we now find the spectral

decomposition:

$$|A - \lambda I| = \begin{vmatrix} 11-\lambda & -3 \\ -3 & 3-\lambda \end{vmatrix} = \lambda^2 - 14\lambda + 24$$

$$= (\lambda - 12)(\lambda - 2)$$

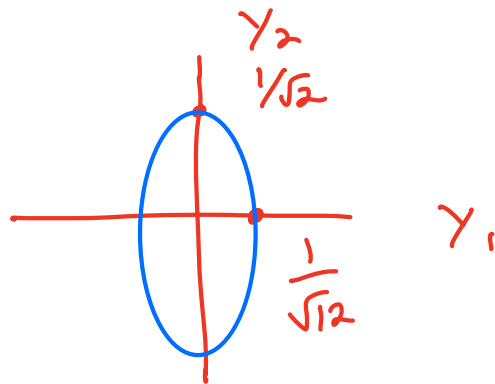
$$\lambda = 12, 2$$

$$\lambda_1 = 12: A - 12I = \begin{bmatrix} -1 & -3 \\ -3 & -9 \end{bmatrix}, \quad \vec{e}_1 = \frac{1}{\sqrt{10}} \begin{bmatrix} 3 \\ -1 \end{bmatrix}$$

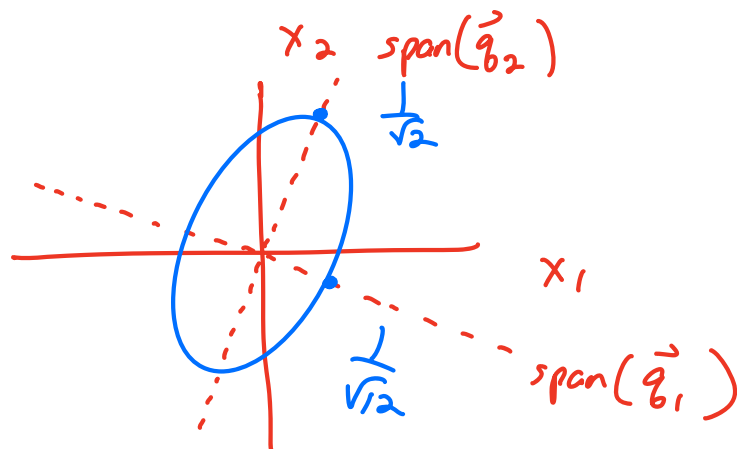
$$\lambda_2 = 2: A - 2I = \begin{bmatrix} 9 & -3 \\ -3 & 1 \end{bmatrix}, \quad \vec{q}_2 = \frac{1}{\sqrt{10}} \begin{bmatrix} 1 \\ 3 \end{bmatrix}$$

As A is pos. def., $\vec{x}^T A \vec{x} = 1$ is an ellipse. Now in the y_1, y_2 axis where $\vec{y} = Q^T \vec{x}$ we have that

$$1 = 12y_1^2 + 2y_2^2$$



Now rotating by Q we obtain the sketch in the x_1, x_2 axis



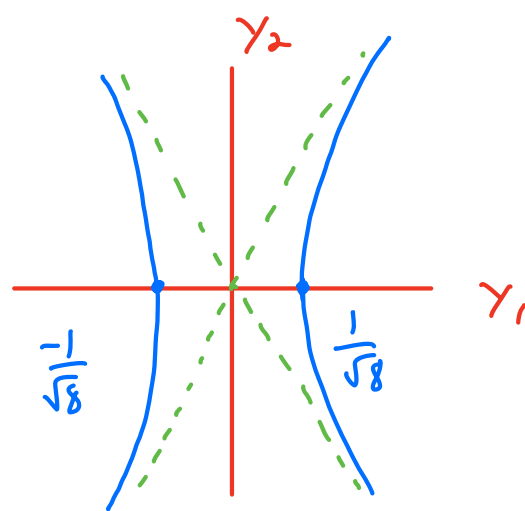
b. $6x_1^2 + 8x_1x_2 = 1$

As $\begin{bmatrix} 6 & 4 \\ 4 & 0 \end{bmatrix}$ has e-values of

8 and -2 and e-vectors of

$$\vec{q}_1 = \frac{1}{\sqrt{5}} \begin{bmatrix} 2 \\ 1 \end{bmatrix} \text{ and } \vec{q}_2 = \frac{1}{\sqrt{5}} \begin{bmatrix} -1 \\ 2 \end{bmatrix}$$

we have that the given conic is a hyperbola. In the y_1, y_2 axis where $\vec{y} = Q^T \vec{x}$ we have $8y_1^2 - 2y_2^2 = 1$

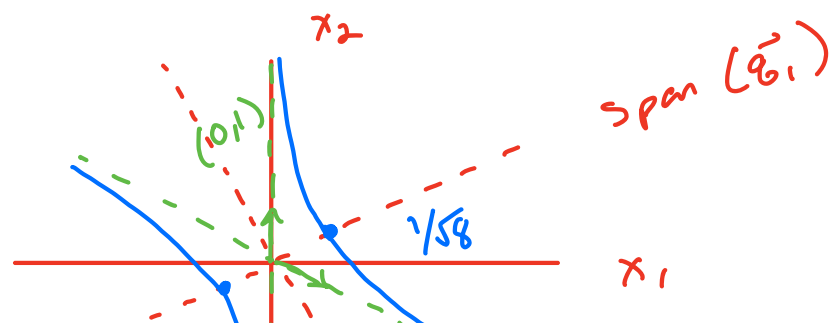


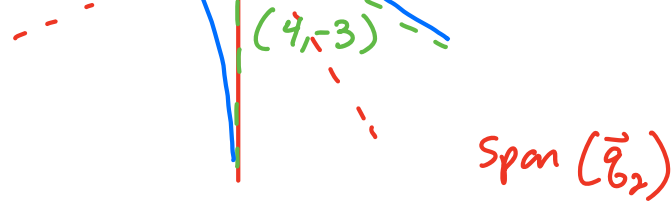
$$\frac{1}{8y_2^2} (8y_1^2 = 1 + 2y_2^2)$$

$$\frac{y_1^2}{y_2^2} = \frac{1}{8y_2^2} + \frac{1}{4}$$

$$\frac{y_1}{y_2} \rightarrow \pm \frac{1}{2}$$

Now rotating by Q





$$Q \begin{bmatrix} 1 \\ 2 \end{bmatrix} = \frac{1}{\sqrt{5}} \begin{bmatrix} 2 & -1 \\ 1 & 2 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \end{bmatrix} \\ = \frac{1}{\sqrt{5}} \begin{bmatrix} 0 \\ 5 \end{bmatrix}$$

$$Q \begin{bmatrix} 1 \\ -2 \end{bmatrix} = \frac{1}{\sqrt{5}} \begin{bmatrix} 2 & -1 \\ 1 & 2 \end{bmatrix} \begin{bmatrix} 1 \\ -2 \end{bmatrix} \\ = \frac{1}{\sqrt{5}} \begin{bmatrix} 4 \\ -3 \end{bmatrix}$$

Singular Value Decomposition (SVD)

21.3 Preliminary Results

i. $\text{null}(AA^*) = \text{null}(A^*)$ and

$$\text{null}(A^*A) = \text{null}(A)$$

ii. AA^* and A^*A have the same non-zero e-values

iii. AA^* and A^*A are both positive semi-definite

Theorem 21.4 Let $A \in \mathbb{C}^{m \times n}$ have rank r .

Then there exists orthonormal bases $\{\vec{u}_1, \dots, \vec{u}_m\}$ and $\{\vec{v}_1, \dots, \vec{v}_n\}$ of $\mathbb{C}^m$ and $\mathbb{C}^n$ respectively and positive real numbers $\sigma_1, \dots, \sigma_r$ such that

$$A \vec{v}_i = \begin{cases} \sigma_i \vec{u}_i & \text{if } i = 1, \dots, r \\ \vec{0} & \text{if } i = r+1, \dots, n \end{cases}$$

Proof: Let $A \in \mathbb{C}^{m \times n}$ and since $AA^* \in \mathbb{C}^{m \times m}$ and $A^*A \in \mathbb{C}^{n \times n}$ are both positive, semi-definite with the same non-zero e-values, let $\{\vec{u}_1, \dots, \vec{u}_m\}$ and $\{\vec{v}_1, \dots, \vec{v}_n\}$ be orthonormal bases s.t. e-vectors of AA^*

$\{\vec{v}_1, \dots, \vec{v}_n\}$ be orthonormal bases s.t. e-vectors of A^*A

$$AA^* = \sum_{i=1}^r \sigma_i^2 \vec{u}_i \vec{u}_i^* \text{ and}$$

$$A^*A = \sum_{i=1}^r \sigma_i^2 \vec{v}_i \vec{v}_i^* . \text{ Now for } 1 \leq i \leq r$$

$$\begin{aligned} AA^*(A \vec{v}_i) &= A(A^*A \vec{v}_i) \\ &= A(\sigma_i^2 \vec{v}_i) \end{aligned}$$

$$= \sigma_i^2 A \vec{v}_i$$

we have that $A \vec{v}_i$ is in the direction of $\vec{u}_i$. Now setting $A \vec{v}_i = \lambda_i \vec{u}_i$ we have that

$$\begin{aligned} \lambda_i^2 &= \lambda_i^2 \langle \vec{u}_i, \vec{u}_i \rangle \\ &= \langle \lambda_i \vec{u}_i, \lambda_i \vec{u}_i \rangle \\ &= \langle A \vec{v}_i, A \vec{v}_i \rangle \\ &= \langle \vec{v}_i, A^* A \vec{v}_i \rangle \\ &= \langle \vec{v}_i, \sigma_i^2 \vec{v}_i \rangle \\ &= \sigma_i^2 \langle \vec{v}_i, \vec{v}_i \rangle = \sigma_i^2 \end{aligned}$$

and thus $\lambda_i = \sigma_i$.

Def 21.5 (SVD) let $\{\vec{u}_1, \dots, \vec{u}_m\}$,

$\{\vec{v}_1, \dots, \vec{v}_n\}$ and $\{\sigma_1, \dots, \sigma_r\}$ be as

above. (I.e. $A \vec{v}_i = \begin{cases} \sigma_i \vec{u}_i & i \leq r \\ \vec{0} & i > r \end{cases}$)

Then $\sigma_1, \dots, \sigma_r$ are the singular values of

A and the SVD is given by

$$A_{m \times n} = U \Sigma V^*$$

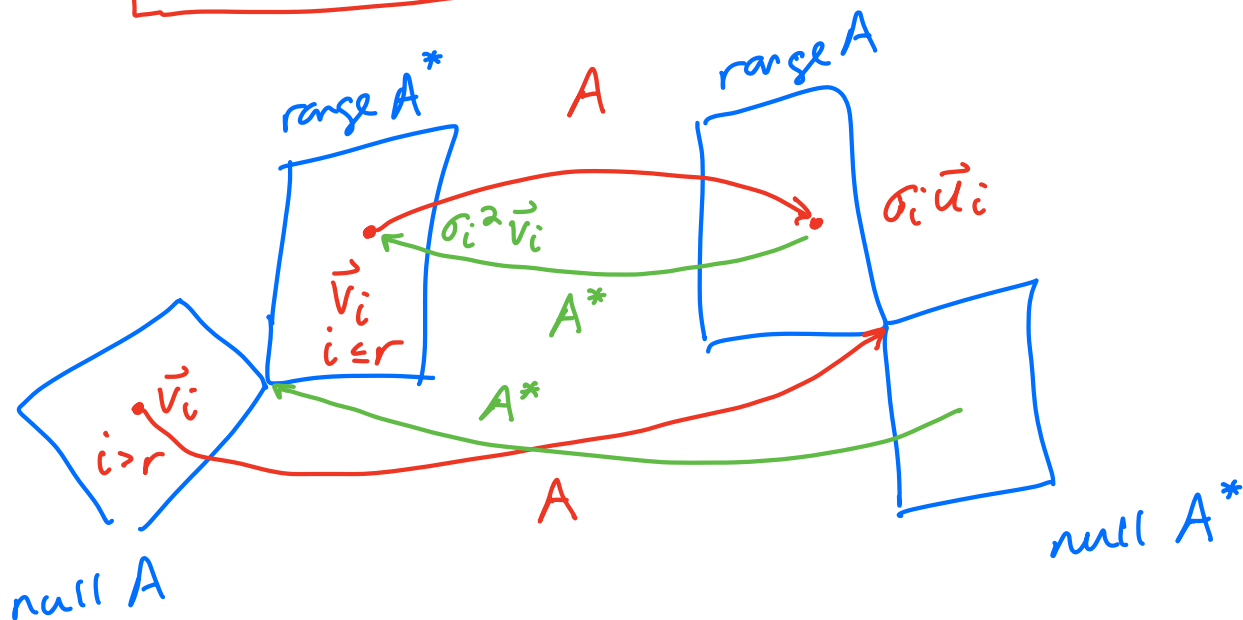
(V and U are both unitary)

where $U = [\vec{u}_1, \dots, \vec{u}_m]_{m \times m}$, $V = [\vec{v}_1, \dots, \vec{v}_n]_{n \times n}$

and $\Sigma = \begin{bmatrix} \sigma_1 & \dots & \sigma_r \\ & & \end{bmatrix}_{m \times n}$. The reduced

SVD of A is given by

$$A = \sum_{i=1}^r \sigma_i \vec{u}_i \vec{v}_i^*$$



Exercise:

$\vec{u}_i = \frac{1}{\sigma_i} A \vec{v}_i$ where $\vec{v}_i$ are the e-vectors of A^*A

3) Find the SVD of the matrices

a. $A = \begin{bmatrix} 3 & 0 \\ 4 & 5 \end{bmatrix}$

Diagonalizing $A^T A = \begin{bmatrix} 3 & 4 \\ 0 & 5 \end{bmatrix} \begin{bmatrix} 3 & 0 \\ 4 & 5 \end{bmatrix}$
 $= \begin{bmatrix} 25 & 20 \\ 20 & 25 \end{bmatrix}$

$$\begin{aligned} |A^T A - \lambda I| &= (25 - \lambda)^2 - 400 \\ &= \lambda^2 - 50\lambda + 225 \\ &= (\lambda - 45)(\lambda - 5) \end{aligned}$$

$$\lambda_1 = 45: A^T A - 45 I = \begin{bmatrix} -20 & 20 \\ 20 & -20 \end{bmatrix}$$

$$\vec{v}_1 = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

$$\lambda_2 = 5: A^T A - 5 I = \begin{bmatrix} 20 & 20 \\ 20 & 20 \end{bmatrix}$$

$$\vec{v}_2 = \frac{1}{\sqrt{2}} \begin{bmatrix} -1 \\ 1 \end{bmatrix}$$

Now setting $\vec{u}_i = \frac{1}{\sigma_i} A \vec{v}_i$

$$\vec{u}_1 = \frac{1}{\sqrt{45}} A \vec{v}_1 = \frac{1}{3\sqrt{5}} \frac{1}{\sqrt{2}} \begin{bmatrix} 3 & 0 \\ 4 & 5 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

$$= \frac{1}{3\sqrt{10}} \begin{bmatrix} 3 \\ 9 \end{bmatrix} = \frac{1}{\sqrt{10}} \begin{bmatrix} 1 \\ 3 \end{bmatrix}$$

$$\begin{aligned} \vec{u}_2 &= \frac{1}{\sqrt{5}} A \vec{v}_2 = \frac{1}{\sqrt{5}} \cdot \frac{1}{\sqrt{2}} \begin{bmatrix} 3 & 0 \\ 4 & 5 \end{bmatrix} \begin{bmatrix} -1 \\ 1 \end{bmatrix} \\ &= \frac{1}{\sqrt{10}} \begin{bmatrix} -3 \\ 1 \end{bmatrix} \end{aligned}$$

$$\therefore A = U \Sigma V^T = \frac{1}{\sqrt{10}} \begin{bmatrix} 1 & -3 \\ 3 & 1 \end{bmatrix} \begin{bmatrix} \sqrt{45} & \\ & \sqrt{5} \end{bmatrix} \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ -1 & 1 \end{bmatrix}$$

or

$$\begin{aligned} A &= \sqrt{45} \vec{u}_1 \vec{v}_1^T + \sqrt{5} \vec{u}_2 \vec{v}_2^T \\ &= \frac{\sqrt{45}}{\sqrt{20}} \begin{bmatrix} 1 \\ 3 \end{bmatrix} \begin{bmatrix} 1 & 1 \end{bmatrix} + \frac{\sqrt{5}}{\sqrt{20}} \begin{bmatrix} -3 \\ 1 \end{bmatrix} \begin{bmatrix} -1 & 1 \end{bmatrix} \\ &= \frac{3}{2} \begin{bmatrix} 1 & 1 \\ 3 & 3 \end{bmatrix} + \frac{1}{2} \begin{bmatrix} 3 & -3 \\ -1 & 1 \end{bmatrix} \end{aligned}$$

$$b. \quad A = \begin{bmatrix} 3 & -1 \\ -4 & -2 \\ 1 & 3 \end{bmatrix}$$

$$A^T A = \begin{bmatrix} 26 & 8 \\ 8 & 14 \end{bmatrix} \quad \lambda = 30, 10$$

$$\vec{v}_1 = \frac{1}{\sqrt{5}} \begin{bmatrix} 2 \\ 1 \end{bmatrix}, \quad \vec{v}_2 = \frac{1}{\sqrt{5}} \begin{bmatrix} -1 \\ 2 \end{bmatrix}$$

$$\begin{aligned}\vec{u}_1 &= \frac{1}{\sqrt{30}} A \vec{v}_1 = \frac{1}{\sqrt{30}} \frac{1}{\sqrt{5}} \begin{bmatrix} 3 & -1 \\ -4 & -2 \\ 1 & 3 \end{bmatrix} \begin{bmatrix} 2 \\ 1 \end{bmatrix} \\ &= \frac{1}{\sqrt{6}} \frac{1}{5} \begin{bmatrix} 5 \\ -10 \\ 5 \end{bmatrix} = \frac{1}{\sqrt{6}} \begin{bmatrix} 1 \\ -2 \\ 1 \end{bmatrix}\end{aligned}$$

$$\begin{aligned}\vec{u}_2 &= \frac{1}{\sqrt{10}} A \vec{v}_2 = \frac{1}{\sqrt{10}} \frac{1}{\sqrt{5}} \begin{bmatrix} 3 & -1 \\ -4 & -2 \\ 1 & 3 \end{bmatrix} \begin{bmatrix} -1 \\ 2 \end{bmatrix} \\ &= \frac{1}{\sqrt{2}} \frac{1}{5} \begin{bmatrix} -5 \\ 0 \\ 5 \end{bmatrix} = \frac{1}{\sqrt{2}} \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix}\end{aligned}$$

$$\vec{u}_3 = \vec{u}_1 \times \vec{u}_2 = \frac{1}{\sqrt{3}} \begin{bmatrix} -1 \\ -1 \\ -1 \end{bmatrix} \quad (\text{sign is irrelevant})$$

$$\therefore A = U \Sigma V^T = \frac{1}{\sqrt{6}} \begin{bmatrix} 1 & -\sqrt{3} & -\sqrt{2} \\ -2 & 0 & -\sqrt{2} \\ 1 & \sqrt{3} & -\sqrt{2} \end{bmatrix} \begin{bmatrix} \sqrt{30} & 0 \\ 0 & \sqrt{10} \\ 0 & 0 \end{bmatrix} \frac{1}{\sqrt{5}} \begin{bmatrix} 2 & 1 \\ -1 & 2 \end{bmatrix}$$

$$= \sqrt{30} \vec{u}_1 \vec{v}_1^T + \sqrt{10} \vec{u}_2 \vec{v}_2^T \quad (\text{reduced SVD})$$

$$= \begin{bmatrix} 1 \\ -2 \\ 1 \end{bmatrix} \begin{bmatrix} 2 & 1 \end{bmatrix} + \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix} \begin{bmatrix} -1 & 2 \end{bmatrix}$$

$$= \begin{bmatrix} 2 & 1 \\ -4 & -2 \\ 2 & 1 \end{bmatrix} + \begin{bmatrix} 1 & -2 \\ 0 & 0 \\ -1 & 2 \end{bmatrix}$$